



Chapter 8. Chemical Equilibrium



Summary

1. Extent of reaction ξ

$$(dG)_{T,p} = \sum_{i=1}^k \mu_i dn_i$$
$$dn_i = \nu_i d\xi$$
$$d\xi = \frac{dn_i}{\nu_i}$$

2. The direction of chemical reactions

T, p constant, $W'=0$

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{T,p} = \sum_{i=1}^k \nu_i \mu_i \leq 0$$



$$\mu_i = \mu_i^\ominus + RT \ln a_i$$

- 化学反应是随着反应进行，体系内组成一直在动态变化的复杂体系。

$$(dG)_{T,p} = \sum_{i=1}^k \mu_i dn_i$$

- 但各个组成的变化不是没有内在联系的随意变化，而是有化学计量关系的变化

$$dn_i = \nu_i d\xi \quad d\xi = \frac{dn_i}{\nu_i}$$

ν_i : Stoichiometry number

ξ : Extent of reaction (moles). 反应进行程度的表征

- The reaction Gibbs energy (J/mol)

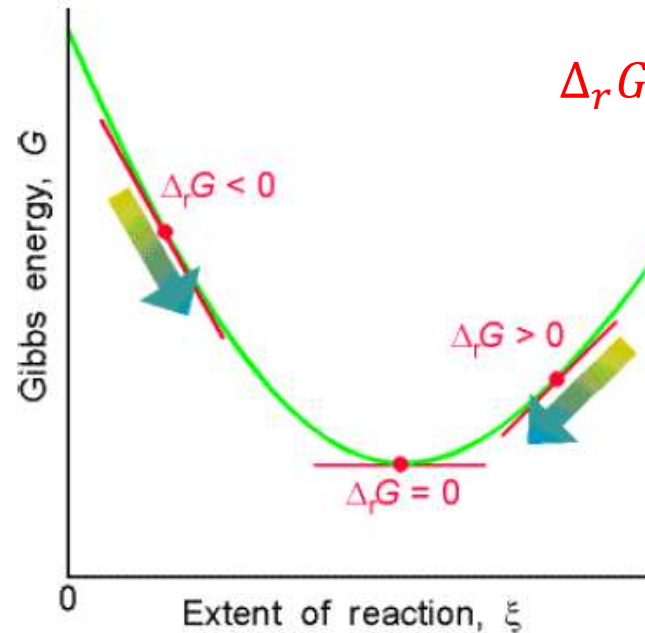
$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{T,p} = \sum_{i=1}^k \nu_i \mu_i \leq 0 \quad \Rightarrow \quad \Delta_r G = \Delta_r G^\ominus(T) + RT \ln Q$$



- $\Delta_r G$

At given T, p

$$\Delta_r G = f(\xi) = f(\text{composition})$$



- ξ 发生变化时，体系的总 $chemical\ potential$ (即吉布斯自由能)不再发生变化就是反应平衡的点。
- 如果 ξ 变化的结果是吉布斯自由能的进一步减小，反应是自发发生的；
- 如果 ξ 变化的结果是吉布斯自由能的增大，逆反应是自发发生的。



3. Isothermal equation of reaction

$$\Delta_r G = \Delta_r G^\ominus(T) + RT \ln Q \quad Q = \prod_B (a_B)^{\nu_B}$$

$$\Delta_r G^\ominus = -RT \ln K \quad K = \left\{ \prod_B (a_B)^{\nu_B} \right\}_{equilibrium}$$

$$\Delta_r G = -RT \ln K + RT \ln Q \quad \left\{ \begin{array}{l} \text{If } Q > K, \Delta_r G > 0 \\ \text{If } Q < K, \Delta_r G < 0 \\ \text{If } Q = K, \Delta_r G = 0 \end{array} \right.$$



- $\Delta_r G^\ominus$

- ✓ 反应物和产物都在标准状态下
- ✓ 温度的函数 $\Delta_r G^\ominus = f(T)$
- ✓ 算法

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$

$$\Delta_r G^\ominus = \sum_{\text{products}} \nu \Delta_f G_m^\ominus - \sum_{\text{reactants}} \nu \Delta_f G_m^\ominus$$

$$\Delta_r G^\ominus = -RT \ln K$$



- Q

- ✓ The reaction quotient measures the relative amounts of products and reactants present during a reaction at a particular point in time.

$$Q = \prod_{B(g)} \left(\frac{f_B}{p^\ominus} \right)^{\nu_B} \cdot \prod_{C(sln)} (a_C)^{\nu_C}$$



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4. Thermodynamic equilibrium constant

$$K = \left\{ \prod_B (a_B)^{\nu_B} \right\}_{\text{equilibrium}}$$

For perfect gas

$$K = \left\{ \prod_B \left(\frac{p_B}{p^\ominus} \right)^{\nu_B} \right\}_{\text{equilibrium}}$$

For solutions

$$K = \left\{ \prod_B (a_B)^{\nu_B} \right\}_{\text{equilibrium}}$$

• K Equilibrium constant

✓ $K = \{Q\}_{equilibrium}$

✓ Dimensionless 无量纲

✓ $K = f(T)$ $K = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$ 温度的函数

✓ 可以判断反应方向: isothermal equation of reaction

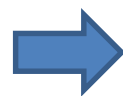
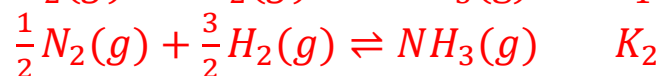
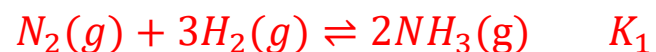
$$\Delta_r G = -RT \ln K + RT \ln Q$$

If $Q > K$, $\Delta_r G > 0$

If $Q < K$, $\Delta_r G < 0$

If $Q = K$, $\Delta_r G = 0$

✓ Determined by the reaction equations



$$K_1 = (K_2)^2$$

✓ 计算

$$\textcircled{1} \quad K = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$$

$$\textcircled{2} \quad K = \left\{ \prod_i \left(\frac{p_i}{p^\ominus}\right)^{v_i} \right\}_{equilibrium}$$



5. Le Chatelier's principle---equilibrium shift

Temperature

$$\frac{d\ln K}{d(1/T)} = -\frac{\Delta_r H^\ominus}{R}$$

If $\Delta_r H^\ominus < 0$; *Exothermic reaction*
 $T \uparrow \quad K \downarrow \quad \text{Shift Left}$

If $\Delta_r H^\ominus > 0$; *Endothermic reaction*
 $T \uparrow \quad K \uparrow \quad \text{Shift Right}$

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Pressure

$$K = f(T) \quad K = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$$

The value of K doesn't change; but the position of equilibrium moves.

- Condition I

System $V \downarrow p \uparrow$

$$K = K_x \cdot \left(\frac{p}{p^\ominus}\right)^{\sum \nu_B}$$

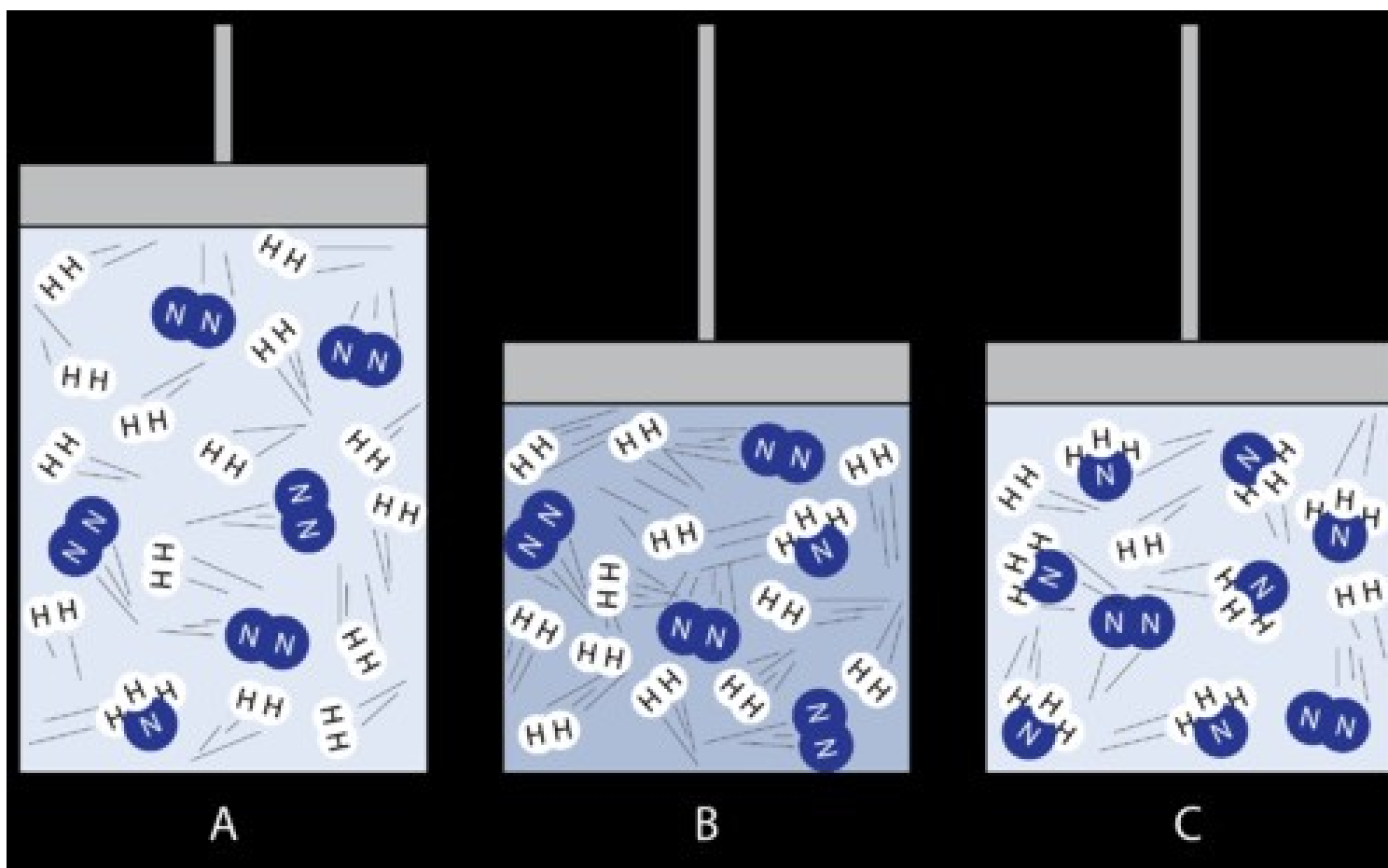
If $\sum \nu_B > 0$;	$K_x \downarrow$	<i>Shift Left</i>
If $\sum \nu_B < 0$;	$K_x \uparrow$	<i>Shift Right</i>
If $\sum \nu_B = 0$;	$K_x \rightarrow$	<i>keep equilibrium</i>

- Condition II

System $V \rightarrow p \uparrow$ (inducing inert gas) *No movement*



Condition I

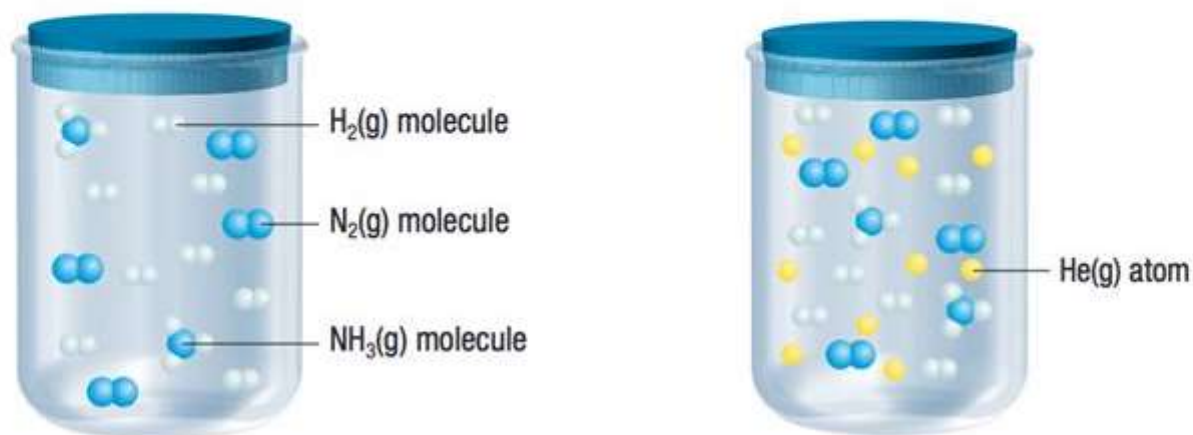




Condition II

- **Inert Gas** – adding an inert gas will increase the pressure thus increasing number of collisions but since it is inert the collisions will not cause reactions leaving the equilibrium intact

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$$K \rightleftharpoons \Delta_r G^\ominus$$

Exercises 9.7, 9.8, 9.9, 9.10, 9.16, 9.19

$$\textcircled{1} \quad K = \left\{ \prod_B \left(\frac{f_B}{p^\ominus} \right)^{\nu_B} \right\}_{\text{equilibrium}} \quad \Delta_r G^\ominus = -RT \ln K$$

$$\textcircled{2} \quad \Delta_r G^\ominus = \sum_{\text{products}} \nu \Delta_f G_m^\ominus - \sum_{\text{reactants}} \nu \Delta_f G_m^\ominus$$

$$K = \exp \left(-\frac{\Delta_r G^\ominus}{RT} \right)$$

$$\textcircled{3} \quad \ln K_2 - \ln K_1 = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T \Delta_r S^\ominus$$



Exercises 9.8

Dinitrogen tetroxide is 18.46 per cent dissociated at 25°C and 1.00 bar in the equilibrium



Calculate:

(a) K at 25°C,

(b) $\Delta_r G^\ominus$,

(c) K at 100°C given that $\Delta_r H^\ominus = +57.2 \text{ kJ mol}^{-1}$ over the temperature range.



Answer for (a):



initial	n	0
reacted	$-\alpha n$	$+2\alpha n$
Final in equilibrium	$(1-\alpha)n$	$+2\alpha n$

$$(x_A)_{eq} = \frac{(1-\alpha)n}{(1-\alpha)n + 2\alpha n} = \frac{1-\alpha}{1+\alpha} \quad (x_B)_{eq} = \frac{2\alpha}{1+\alpha}$$

$$K = \frac{f_B^2}{f_A} = \frac{x_B^2 p}{x_A} = \frac{4\alpha^2}{1-\alpha^2} \frac{p}{p^\ominus}$$

$$p = p^\ominus \quad \alpha = 0.1846$$

$$K = 0.141$$



Answer for (b):

$$\Delta_r G^\ominus = -RT \ln K = -8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \times 298 \text{ K} \times \ln 0.141$$

$$\Delta_r G^\ominus = 4.86 \times 10^3 \text{ J mol}^{-1}$$

Answer for (c):

$$\ln K_{373\text{K}} - \ln K_{298\text{K}} = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln K_{373\text{K}} = \ln 0.141 - \frac{5.72 \times 10^4 \text{ J mol}^{-1}}{8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}} \left(\frac{1}{373\text{K}} - \frac{1}{298\text{K}} \right)$$

$$\ln K_{373\text{K}} = 14.556$$



Exercises 9.9

From information in the *Data section*, calculate the standard Gibbs energy and the equilibrium constant at (a) 298 K and (b) 400 K for the reaction



Assume that the reaction enthalpy is independent of temperature.



$$\Delta_r G^\ominus = \Delta_f G_m^\ominus(\text{CO}_2, g) - \Delta_f G_m^\ominus(\text{CO}, g) - \Delta_f G_m^\ominus(\text{PbO}, s)$$

$$\Delta_r G^\ominus (298 \text{ K})$$

$$= \{-3.9436 \times 10^5 - (-1.3717 \times 10^5) - (-1.8893 \times 10^5)\} J \text{ mol}^{-1}$$

$$\Delta_r G^\ominus (298 \text{ K}) = -6.826 \times 10^4 J \text{ mol}^{-1}$$

$$K(298 \text{ K}) = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right) = 9.2 \times 10^{11}$$



$$\Delta_r H^\ominus = \Delta_f H_m^\ominus(\text{CO}_2, g) - \Delta_f H_m^\ominus(\text{CO}, g) - \Delta_f H_m^\ominus(\text{PbO}, s)$$

$$\begin{aligned} \Delta_r H^\ominus(298\text{ K}) \\ = \{-3.9351 \times 10^5 - (-1.1053 \times 10^5) - (-2.1899 \times 10^5)\} J \text{ mol}^{-1} \end{aligned}$$

$$\Delta_r H^\ominus(298\text{ K}) = -6.399 \times 10^4 J \text{ mol}^{-1}$$

$$\ln K_{400\text{ K}} - \ln K_{298} = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln K_{400\text{ K}} = \ln 9.2 \times 10^{11} + \frac{6.399 \times 10^4 J \text{ mol}^{-1}}{8.314 J \cdot K^{-1} \cdot \text{mol}^{-1}} \left(\frac{1}{400\text{ K}} - \frac{1}{298\text{ K}} \right)$$

$$K_{400\text{ K}} = 1.27 \times 10^9 J \text{ mol}^{-1}$$

$$\Delta_r G^\ominus(400\text{ K}) = -RT \ln K = -8.314 J \cdot K^{-1} \cdot \text{mol}^{-1} \times 400\text{ K} \times 20.964$$

$$\Delta_r G^\ominus = -6.97 \times 10^4 J \text{ mol}^{-1}$$



9.10 a

7.4(a) In the gas-phase reaction $2A + B \rightleftharpoons 3C + 2D$, it was found that, when 1.00 mol A, 2.00 mol B, and 1.00 mol D were mixed and allowed to come to equilibrium at 25°C, the resulting mixture contained 0.90 mol C at a total pressure of 1.00 bar. Calculate (a) the mole fractions of each species at equilibrium, (b) K_x , (c) K , and (d) $\Delta_r G^\ominus$.



	A	B	C	D
ν	-2	-1	3	2
Initial	1 mol	2 mol	0 mol	1 mol
Equi	0.4 mol	1.7 mol	0.9 mol	1.6 mol
x_i mol fraction	0.087	0.370	0.196	0.348

$$n_{\text{overall}} = 4.6 \text{ mol}$$

$$d\xi = \frac{dn_i}{\nu_i} = 0.3 \text{ mol}$$

$$dn_A = -0.6 \text{ mol} \quad dn_B = -0.3 \text{ mol} \quad dn_D = 0.6 \text{ mol}$$



$$K = K_x \cdot \left(\frac{p}{p^\ominus}\right)^{\sum \nu_B}$$

$$p = p^\ominus$$

$$K = K_x = \frac{x_C^3 \cdot x_D^2}{x_A^2 \cdot x_B} = \frac{0.196^3 \times 0.348^2}{0.087^2 \times 0.370} = 0.3256$$

$$\begin{aligned}\Delta_r G^\ominus &= -RT \ln K = -8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \times 298 \text{ K} \times \ln 0.3256 \\ &= 2.72 \times 10^3 \text{ J mol}^{-1}\end{aligned}$$



7.11(b) The dissociation vapour pressure of NH_4Cl at 427°C is 608 kPa but at 459°C it has risen to 1115 kPa. Calculate (a) the equilibrium constant, (b) the standard reaction Gibbs energy, (c) the standard enthalpy, (d) the standard entropy of dissociation, all at 427°C . Assume that the vapour behaves as a perfect gas and that $\Delta H^\ominus$ and $\Delta S^\ominus$ are independent of temperature in the range given.



$$p = p_{\text{NH}_3} + p_{\text{HCl}}$$

$$\because p_{\text{NH}_3} = p_{\text{HCl}} \quad \therefore p_{\text{NH}_3} = p_{\text{HCl}} = \frac{1}{2}p$$

$$K = \frac{p_{\text{NH}_3}}{p^\ominus} \times \frac{p_{\text{HCl}}}{p^\ominus} = \frac{1}{4} \times \frac{p^2}{(p^\ominus)^2}$$

$$K_{700} = \frac{1}{4} \times \left(\frac{608000 \text{ Pa}}{100000 \text{ Pa}} \right)^2 = 9.24$$

$$K_{732} = \frac{1}{4} \times \left(\frac{1115000 \text{ Pa}}{100000 \text{ Pa}} \right)^2 = 31.08$$



$$\begin{aligned}\Delta_r G^\ominus &= -RT \ln K = -8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \times 700 \text{ K} \times \ln 9.24 \\ &= -1.294 \times 10^4 \text{ Jmol}^{-1}\end{aligned}$$

$$\ln K_{732\text{K}} - \ln K_{700\text{K}} = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln 31.08 - \ln 9.24 = -\frac{\Delta_r H^\ominus}{8.314} \left(\frac{1}{732} - \frac{1}{700} \right)$$

$$\Delta_r H^\ominus = 1.615 \times 10^5 \text{ Jmol}^{-1}$$

$$\begin{aligned}\Delta_r S^\ominus &= \frac{\Delta_r H^\ominus - \Delta_r G^\ominus}{T} = \frac{1.615 \times 10^5 \text{ Jmol}^{-1} + 1.294 \times 10^4 \text{ Jmol}^{-1}}{700 \text{ K}} \\ &= 249.2 \text{ Jmol}^{-1} \text{ K}^{-1}\end{aligned}$$



7.5(b) The standard enthalpy of a certain reaction is approximately constant at $+125 \text{ kJ mol}^{-1}$ from 800 K up to 1500 K. The standard reaction Gibbs energy is $+22 \text{ kJ mol}^{-1}$ at 1120 K. Estimate the temperature at which the equilibrium constant becomes greater than 1.

$$\Delta_r G^\ominus_{1120\text{K}} = 22 \text{ kJ mol}^{-1} \xrightarrow{\text{可求}} K_{1120\text{K}} = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$$

$\Delta_r H^\ominus > 0$ Endothermic reaction, if $T \uparrow$, *shift to right*

$$\ln K - \ln K_{1120\text{K}} = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T} - \frac{1}{1120} \right)$$

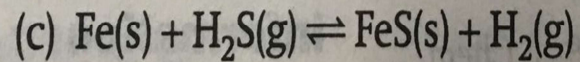
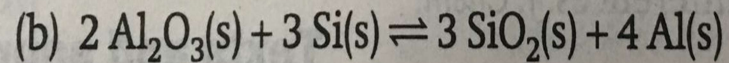
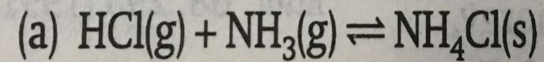
$$\Delta_r H^\ominus = 125 \text{ kJ mol}^{-1}$$

$$K = 1 \xrightarrow{\quad} \ln K = 0 \quad \text{可求此时的温度 } T$$



9.16 (a) Use the data in Table 2.6 of the *Data section* to decide which of the following reactions have $K > 1$ at 298 K.

d

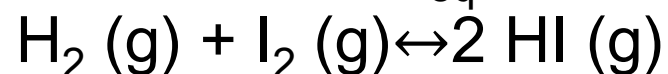


$$K > 1 \quad \Rightarrow \quad \Delta_r G^\ominus = -RT \ln K < 0$$

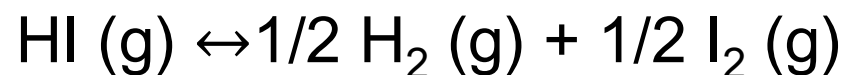
根据书后的表格计算 $\Delta_r G^\ominus$



The value of K_{eq} for the equilibrium



is 794 at 25°C. At this temperature, what is the value of K_{eq} for the equilibrium below?

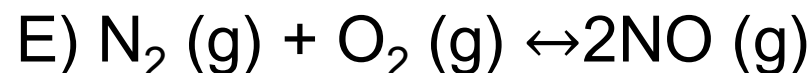
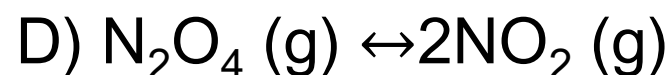
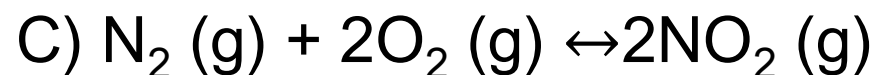
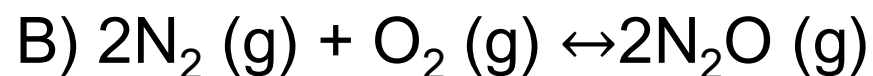
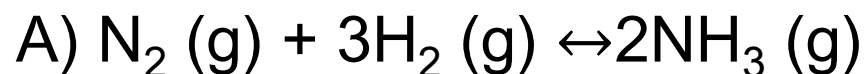


- A) 0.035
- B) 0.0013
- C) 28
- D) 397
- E) 1588

A



In which of the following reactions would increasing pressure at constant temperature not change the concentrations of reactants and products?



E



Chapter 9. Equilibrium Electrochemistry

Summary

1. Standard functions of formation of ions

Exercises 10.4

$$\Delta_f H^\ominus(H^+, aq) = 0 \quad \Delta_f G^\ominus(H^+, aq) = 0$$
$$S^\ominus(H^+, aq) = 0$$

2. Mean activity coefficients (Debye-Hückel limiting law)

Exercises 10.7, 10.8 and 10.12

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2} \quad I = \frac{1}{2} \sum_i z_i^2 (b_i / b^\ominus)$$

从离子氛的模型推导而来，求的是一种电解质电离形成的一对阴阳离子的平均活度修正因子

z_+ 阳离子的电荷数； z_- 和阳离子抗衡的阴离子的电荷数

包括溶液中所有离子

$$A = 0.509 \text{ for water medium at } 298K$$



A compound M_pX_q that dissolves to give a solution of p cations and q anions, the mean activity coefficient:

$$\gamma_{\pm} = (\gamma_+^p \gamma_-^q)^{1/s} \quad s = p + q$$

The chemical potential of each ion:

$$\mu_+ = \mu_+^{ideal} + RT \ln \gamma_{\pm}$$

$$\mu_- = \mu_-^{ideal} + RT \ln \gamma_{\pm}$$

The total chemical potential of the ions in the electrically neutral solution:

$$\mu = p\mu_+ + q\mu_- = \mu^{ideal} + sRT \ln \gamma_{\pm}$$



10.7 (a) Relate the ionic strengths of (a) KCl, (b) FeCl₃, and (c) CuSO₄ solutions to their molalities, b .

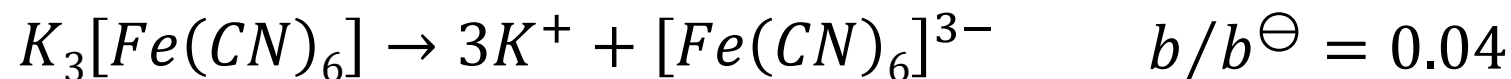
$$I = \frac{1}{2} \sum_i z_i^2 (b_i / b^\ominus)$$



$$I = \frac{1}{2} (3^2 + 1^2 + 1^2 + 1^2) (b / b^\ominus) = 6b / b^\ominus$$



10.8 (b) Calculate the ionic strength of a solution that is $0.040 \text{ mol kg}^{-1}$ in $\text{K}_3[\text{Fe}(\text{CN})_6](\text{aq})$, $0.030 \text{ mol kg}^{-1}$ in $\text{KCl}(\text{aq})$, and $0.050 \text{ mol kg}^{-1}$ in $\text{NaBr}(\text{aq})$.



$$I = \frac{1}{2} (3 \times 1^2 + 3^2) \times 0.04 + \frac{1}{2} (1^2 + 1^2) \times 0.03 + \frac{1}{2} (1^2 + 1^2) \times 0.05 = 0.32$$



Exercises 10.12a

Estimate the mean activity coefficient of a solution that is $0.010 \text{ mol kg}^{-1} \text{ CaCl}_2 \text{ (aq)}$ and $0.030 \text{ mol kg}^{-1} \text{ NaF (aq)}$

$$I = \frac{1}{2} \sum_i z_i^2 (b_i / b^\ominus)$$



包括溶液中所有离子

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2}$$



从离子氛的模型推导而来，求的是一种电解质电离形成的一对阴阳离子的平均活度修正因子。

z_+ 阳离子的电荷数；

z_- 和阳离子抗衡的阴离子的电荷数



	Ca ²⁺	Cl ⁻	Na ⁺	F ⁻
Z _i	+2	-1	+1	-1
b (mol kg ⁻¹)	0.01	2*0.02	0.03	0.03

$$I = \frac{1}{2} \sum_i z_i^2 (b_i / b^\ominus)$$

$$= \frac{1}{2} (2^2 + 2 \times 1^2) \times 0.01 + \frac{1}{2} (1^2 + 1^2) \times 0.03 = 0.06$$

For CaCl₂

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2} = -0.509 \times 2 \times \sqrt{0.06} = -0.249$$

$$\gamma_{\pm}(\text{CaCl}_2) = 0.564$$

For NaF

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2} = -0.509 \times 1 \times \sqrt{0.06} = -0.125$$

$$\gamma_{\pm}(\text{NaF}) = 0.750$$



3. The bridge between electrochemical reactions and thermodynamics. [Exercises 10.22](#)

Constant T and p

$$\Delta_r G = -vFE$$

$$\Delta_r G^\ominus = -vFE^\ominus \quad (\ominus: b_{ion} = 1 \text{ mol/kg})$$

$$E^\ominus = f(T)$$

E: electromotive force, emf (dimension is V).

The ability of galvanic cells to convert the chemical energy to the electrical energy.

We must ensure that the cell is operating reversibly at a specific, constant composition.



4. Half reaction

Exercises 10.18, 10.19

The right-hand electrode: Cathode (Reduction reaction)

The left-hand electrode: Anode (Oxidation reaction)

How to design a galvanic cell?

Compare the reduction ability between two electrodes



$$E = E(\text{right}) - E(\text{left})$$

$$E^\ominus = E^\ominus(\text{right}) - E^\ominus(\text{left})$$

$$E^\ominus\{H^+(aq, b^\ominus)|H_2(g, p^\ominus)\} = 0$$



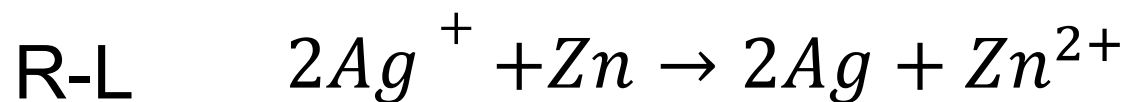
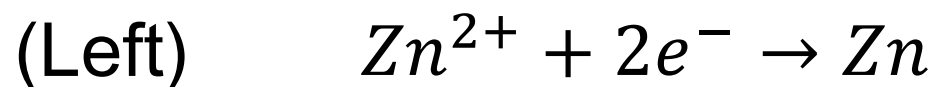
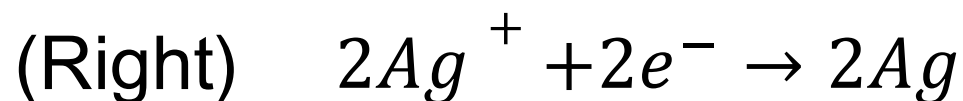
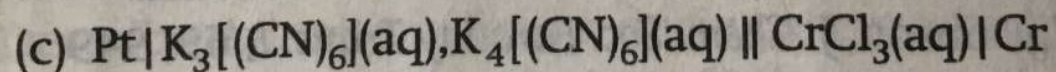
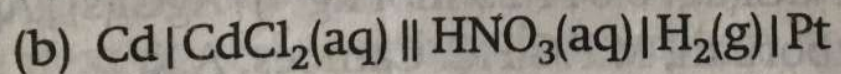
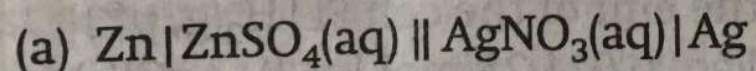
$$E^{\ominus}$$

- The reduction potential of an electrode (a redox couple)
- If a half reaction is multiplied by numerical factor, $E^{\ominus}$ is not affected.
- The values of common electrodes can be obtained from information table 10.7.

$$\ln K = \frac{vFE^{\ominus}}{RT} \quad \text{Calculate equilibrium constant}$$



(10.18 (a)) Write the cell reactions and electrode half-reactions for the following cells:

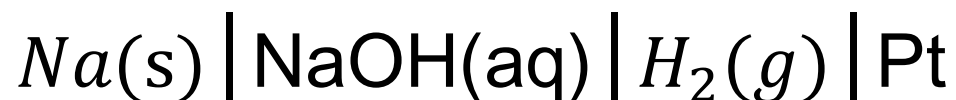
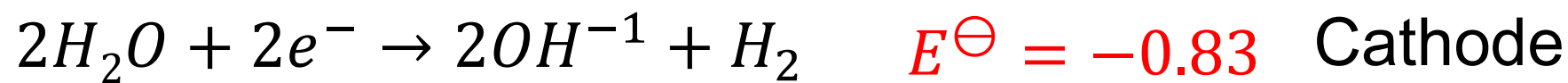
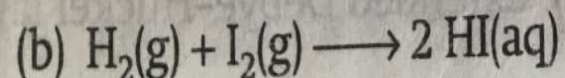
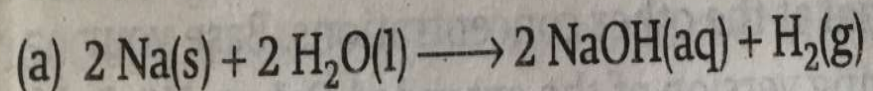


注意：配平！！

Attention: the final cell reaction should be neutral!!



10.19 (b) Devise cells in which the following are the reactions:





5. Nernst equation [Exercises 10.16, 10.24 and 10.26](#)

$$E = E^{\ominus} - \frac{RT}{vF} \ln Q$$

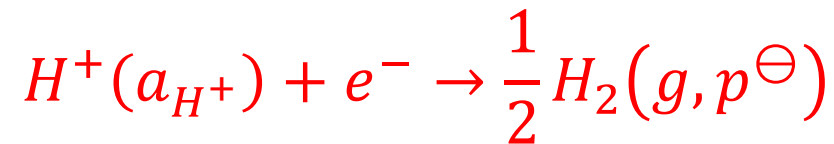
可对应电极反应，此时电子e不计入反应商
也可对应电池的总氧化还原反应

$$emf = f(T, p, composition)$$

$emf = 0$ the cell reach its chemical equilibrium



10.16 (b) Consider a hydrogen electrode in aqueous HCl solution at 25°C operating at 105 kPa. Calculate the change in the electrode potential when the molality of the acid is changed from 5.0 mmol kg⁻¹ to 50 ~~mol~~ mmol kg⁻¹. Activity coefficients are given in Table 10.5.



$$E = E^\ominus - \frac{RT}{vF} \ln Q = -\frac{RT}{F} \ln \frac{(p_{H_2}/p^\ominus)^{1/2}}{a_{H^+}}$$

$$\Delta E = E_2 - E_1 = \frac{RT}{F} \ln \frac{a_{2H^+}}{a_{1H^+}} = \frac{RT}{F} \ln \frac{\gamma_{\pm 2} \times b_2}{\gamma_{\pm 1} \times b_1}$$

$$= 25.7 \text{ mV} \times \ln \frac{0.830 \times 5 \times 10^{-2}}{0.925 \times 5 \times 10^{-3}} = 56.4 \text{ mV}$$



Exercises 10.24

Consider the cell



for which the cell reaction is



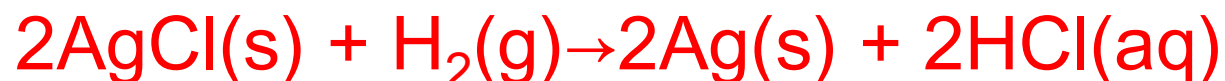
At 25°C and a molality of HCl of 0.010 mol kg⁻¹, $E = +0.4658 \text{ V}$.

- (a) Write the Nernst equation for the cell reaction.
- (b) Calculate $\Delta_r G$ for the cell reaction.
- (c) Assuming that the Debye–Hückel limiting law holds at this concentration, calculate $E^\ominus(\text{AgCl}, \text{Ag})$.



We know:

- ✓ the cell reaction
- ✓ $T=298\text{K}$, $b_{\text{HCl}}=0.01\text{ mol/kg}$; $E=+0.4658\text{ V}$



Answer for (a):

$$E = E^\ominus - \frac{RT}{vF} \ln Q$$

$$E = E^\ominus - \frac{RT}{2F} \ln \frac{(a_{\text{H}^+} \times a_{\text{Cl}^-})^2}{f_{\text{H}_2} / p^\ominus} = E^\ominus - \frac{2RT}{F} \ln \gamma_{\pm} b / b^\ominus$$



Answer for (b):

$$\Delta_r G = -vFE = -2 \times 9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1} \times 0.4658 \text{ V}$$

$$\Delta_r G = -8.99 \times 10^5 \text{ J} \cdot \text{mol}^{-1}$$

Answer for (c):

1. Ions in the system: H^+ and Cl^-

2. $\text{Pt} \mid \text{H}_2(\text{g}, p^\ominus) \mid \text{HCl}(\text{aq}) \mid \text{AgCl}(\text{s}) \mid \text{Ag}$

$$E^\ominus = E^\ominus(\text{AgCl}/\text{Ag}^+, \text{Cl}^{-1}) - E^\ominus(\text{H}^+/\text{H}_2)$$



$$E = E^\ominus - \frac{2RT}{F} \ln \gamma_{\pm} b$$



$$0$$



$$I = \frac{1}{2} \sum_i z_i^2 (b_i / b^\ominus) = \frac{1}{2} (1^2 + 1^2) \times \frac{0.01 \text{ mol} \cdot \text{kg}^{-1}}{1 \text{ mol} \cdot \text{kg}^{-1}}$$

$$I = 0.01$$

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2} = -0.509 \times \sqrt{0.01} = -0.0509$$

$$\gamma_{\pm} = 0.89$$

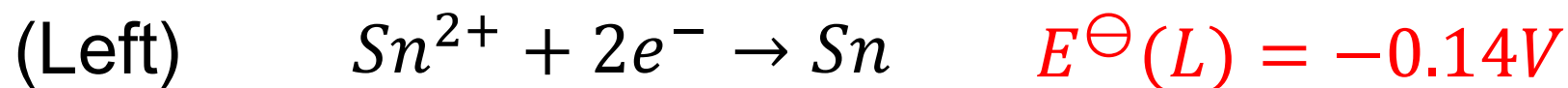
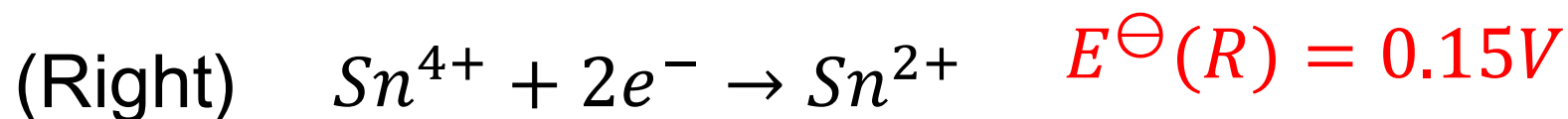
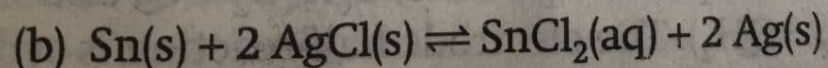
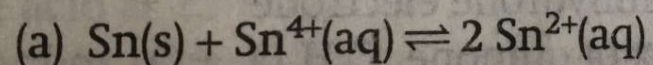
$$E = E^\ominus - \frac{2RT}{F} \ln \gamma_{\pm} b$$

$$\begin{aligned} E^\ominus &= E + \frac{2RT}{F} \ln \gamma_{\pm} b \\ &= 0.4658 \text{ V} + \frac{2 \times 8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \times 298 \text{ K}}{9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1}} \ln 0.89 \times 0.01 \end{aligned}$$

$$E^\ominus (\text{cell}) = E^\ominus (\text{AgCl}/\text{Ag}^+, \text{Cl}^{-1}) = 0.223 \text{ V}$$



10.26 (a) Calculate the equilibrium constants of the following reactions at 25°C from standard potential data:



$$E^\ominus(\text{cell}) = 0.29 \text{ V}$$

$$\ln K = \frac{vFE^\ominus}{RT} = \frac{2 \times 9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1} \times 0.29\text{V}}{8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \times 298\text{K}}$$

$$K = 6.5 \times 10^9$$



6. Calculate thermodynamic functions

Exercises 10.25

$$\Delta_r G^\ominus = -vFE^\ominus$$

$$\frac{dE^\ominus}{dT} = \frac{\Delta_r S^\ominus}{vF}$$

The temperature coefficient of the cell potential

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$



课堂例题

Using the temperature coefficient of the cell potential

The standard cell potential of



was measured over a range of temperatures, and the data were fitted to the following polynomial:

$$E^\ominus/\text{V} = 0.07131 - 4.99 \times 10^{-4}(T/\text{K} - 298) - 3.45 \times 10^{-6}(T/\text{K} - 298)^2$$

Evaluate the standard reaction Gibbs energy, enthalpy, and entropy at 298 K.



Method: The **standard Gibbs energy** of reaction is obtained by using the following equation after evaluating $E^\ominus$ at 298 K.

$$E^\ominus = -\frac{\Delta_r G^\ominus}{\nu F}$$

The **standard entropy** of reaction is obtained by using the following equation, which involves differentiating the polynomial with respect to T and then setting $T = 298$ K.

$$\frac{dE^\ominus}{dT} = \frac{\Delta_r S^\ominus}{\nu F}$$

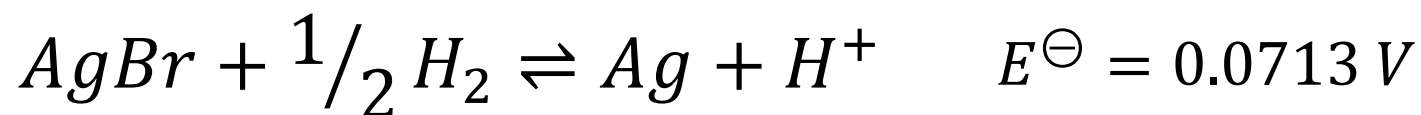
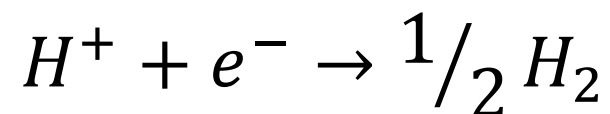
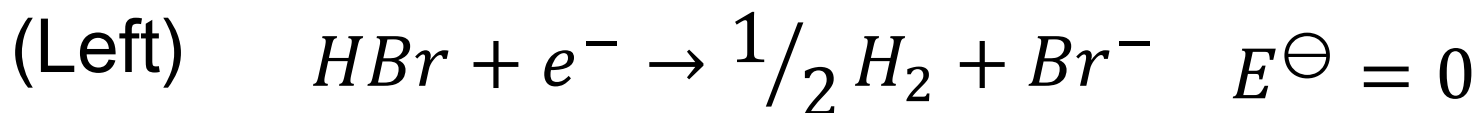
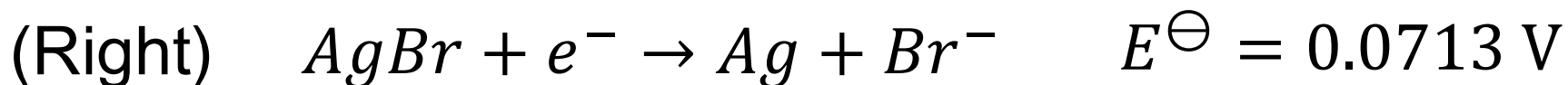
The **reaction enthalpy** is obtained by combining the values of the standard Gibbs energy and entropy.



$$E^{\ominus} = 0.07131 - 4.99 \times 10^{-4}(T - 298) - 3.45 \times 10^{-6}(T - 298)^2$$

$$T = 298 \text{ K}$$

$$E^{\ominus} = 0.07131 \text{ V}$$





$$\begin{aligned}\Delta_r G^\ominus &= -\nu F E^\ominus = -1 \times 9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1} \times 0.07131 \\ &= -6.880 \times 10^3 \text{ Jmol}^{-1}\end{aligned}$$

$$E^\ominus = 0.07131 - 4.99 \times 10^{-4}(T - 298) - 3.45 \times 10^{-6}(T - 298)^2$$

$$\frac{dE^\ominus}{dT} = -4.99 \times 10^{-4} - 2 \times 3.45 \times 10^{-6}(T - 298) = -4.99 \times 10^{-4}$$

$$\frac{dE^\ominus}{dT} = \frac{\Delta_r S^\ominus}{\nu F}$$

$$\begin{aligned}\Delta_r S^\ominus &= \nu F \frac{dE^\ominus}{dT} = 1 \times 9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1} \times (-4.99 \times 10^{-4}) \\ &= -48.2 \text{ J K}^{-1} \text{ mol}^{-1}\end{aligned}$$

$$\begin{aligned}\Delta_r H^\ominus &= \Delta_r G^\ominus + T \Delta_r S^\ominus \\ &= -2.12 \times 10^4 \text{ Jmol}^{-1}\end{aligned}$$



Exercises 10.25

(a) Use the information in Tables 10.7 to calculate the standard emf of the cell:

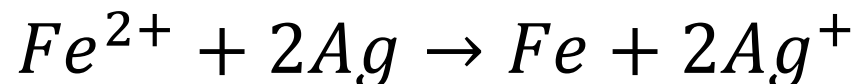
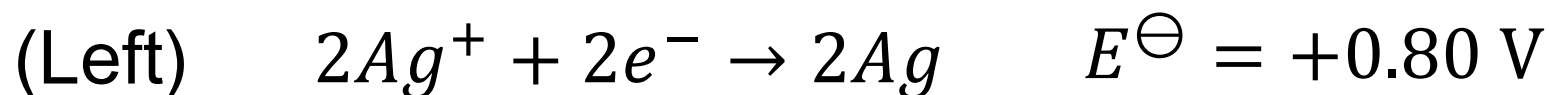
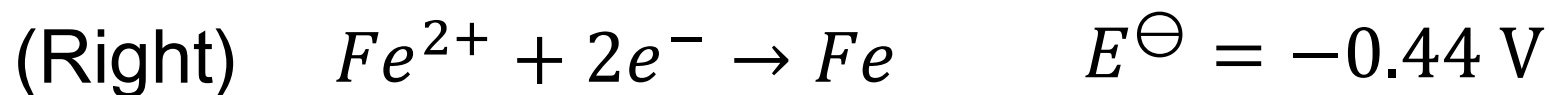


(b) Calculate the standard Gibbs energy and enthalpy of the cell reaction at 25°C.

(c) Estimate the value of $\Delta_r G^\ominus$ at 35 °C



Answer for (a):



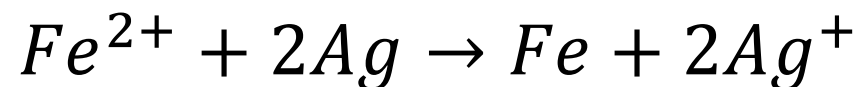
$$E^\ominus = E^\ominus(\text{right}) - E^\ominus(\text{left}) = -1.24 \text{ V}$$



Answer for (b):

$$\Delta_r G^\ominus = -vFE^\ominus = -2 \times 9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1} \times (-1.24 \text{ V})$$

$$\Delta_r G^\ominus(298\text{K}) = 2.39 \times 10^5 \text{ J} \cdot \text{mol}^{-1}$$



$$\begin{aligned} \Delta_r H^\ominus(298\text{K}) &= 2\Delta_f H^\ominus(\text{Ag}^+, \text{aq}) - \Delta_f H^\ominus(\text{Fe}^{2+}, \text{aq}) \\ &= 3.00 \times 10^5 \text{ J} \cdot \text{mol}^{-1} \end{aligned}$$



Answer for (c):

$$\ln K(298K) = \frac{vFE}{RT} = \frac{2 \times 9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1} \times (-1.24 \text{ V})}{8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \times 298 \text{ K}} = -96.6$$

$$\begin{aligned} \ln K(308K) - \ln K(298K) &= -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{308 \text{ K}} - \frac{1}{298 \text{ K}} \right) \\ &= -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{308 \text{ K}} - \frac{1}{298 \text{ K}} \right) = -\frac{3.00 \times 10^5 \text{ J} \cdot \text{mol}^{-1}}{8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}} \left(\frac{1}{308 \text{ K}} - \frac{1}{298 \text{ K}} \right) \\ &= 3.93 \end{aligned}$$

$$\ln K(308K) = -92.67$$

$$\Delta_r G^\ominus(308K) = -RT \ln K(308 \text{ K}) = 2.37 \times 10^5 \text{ J} \cdot \text{mol}^{-1}$$



Table 20.1

Half Reaction	$E^\circ(\text{V})$
$\text{F}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{F}^-(\text{aq})$	+2.87
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$	+1.359
$\text{Br}_2(\text{l}) + 2\text{e}^- \rightarrow 2\text{Br}^-(\text{aq})$	+1.065
$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$	+1.23
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}(\text{s})$	+0.799
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.771
$\text{I}_2(\text{s}) + 2\text{e}^- \rightarrow 2\text{I}^-(\text{aq})$	+0.536
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	0
$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$	-0.126
$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$	-0.28
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.05

Which of the halogens in Table 20.1 is the strongest oxidizing agent?

A) Br_2

B) I_2

C) Cl_2

D) F_2

E) All of the halogens have equal strength as oxidizing agents.

D



Concepts

- Mean activity coefficient
- Ionic strength
- Galvanic cell
- Electrolytic cell
- Oxidation
- Reduction
- Oxidizing agent
- Reducing agent
- The liquid junction potential
- Salt bridge
- Half-reaction
- Electromotive potential
- Concentration cells
- Standard potential
- Standard hydrogen Electrode (SHE)
- The temperature coefficient of the cell potential



Chapter 5

Simple mixtures



Summary

1. Partial molar quantities

➤ definition

$$X_B = \left(\frac{\partial X}{\partial n_B} \right)_{T, p, n_A}$$

$$dX = X_A dn_A + X_B dn_B$$

➤ Additive formula [Exercises 7.4 and 7.5](#)

$$X = X_A n_A + X_B n_B$$

One state at certain T, p, x_A

非常重要

➤ Gibbs-Duhem equation

$$n_A d\mu_A + n_B d\mu_B = 0$$



$$X = X_A n_A + X_B n_B$$

Provided the composition is held constant as the amounts of A and B are increased, the final quantity of a mixture can be calculated by integration. Because the partial molar quantities are constant (provided the composition is held constant throughout the integration) we can write:



The Gibbs-Duhem equation

$$\sum_J n_J d\mu_J = 0$$

The significance of the Gibbs–Duhem equation is that the chemical potential of one component of a mixture cannot change independently of the chemical potentials of the other components.

In a binary mixture, if one partial molar quantity increases, then the other must decrease, with the two changes related by:

$$dV_B = -\frac{n_A}{n_B} dV_A$$



7.4 a

5.1(a) The partial molar volumes of acetone (propanone) and chloroform (trichloromethane) in a mixture in which the mole fraction of CHCl_3 is 0.4693 are $74.166 \text{ cm}^3 \text{ mol}^{-1}$ and $80.235 \text{ cm}^3 \text{ mol}^{-1}$, respectively. What is the volume of a solution of mass 1.000 kg?

Handwritten solution for problem 5.1(a):

$M(\text{acetone}) = 58.08$
 $M(\text{chloroform}) = 119.38$
 $V_a = 74.166 \text{ cm}^3 \text{ mol}^{-1}$
 $V_c = 80.235 \text{ cm}^3 \text{ mol}^{-1}$
 $x_c = 0.4693 \quad x_a = 0.5307$

Formula for volume:

$$V = V_a \times n_a + V_c \times n_c \quad \text{混合公式}$$

Mass calculation:

$$m = n \times x_a \times M_a + n \times x_c \times M_c$$

$$m = n (M_a \cdot x_a + M_c \cdot x_c)$$

$$1 \text{ kg} = n (58.08 \times 0.5307 + 119.38 \times 0.4693)$$

$$n = n (30.823 + 56.025)$$

$$1000 \text{ g} = n \times 86.848 \text{ g/mol}$$

$$n = 11.514 \text{ mol}$$

Moles of each component:

$$n_a = 11.514 \times 0.5307 = 6.11 \text{ mol}$$

$$n_c = 11.514 \times 0.4693 = 5.40 \text{ mol}$$

Final volume calculation:

$$V = V_a \times n_a + V_c \times n_c$$

$$= 74.166 \times 6.11 + 80.235 \times 5.40$$

$$= 453.15 + 433.269$$

$$= 886.42 \text{ cm}^3$$



2. The Gibbs energy of mixing [Self-test 7.2 and 7.3](#) [and Exercise 7.12a](#)

- General gases mixing at constant T

$$\Delta G_{mix} = n_A RT \ln \frac{p_A}{p^*(A)} + n_B RT \ln \frac{p_B}{p^*(B)}$$

- Perfect Gases/ideal solutions mixing at constant T, p and V ($V_A + V_B = V$)

$$\Delta G_{mix} = nRT(x_A \ln x_A + x_B \ln x_B) < 0$$

$$\Delta_{mix} S = -nR(x_A \ln x_A + x_B \ln x_B) > 0$$

$$\Delta_{mix} H = 0$$



Self-test 7.2

A container is divided into two equal compartments. One contains 3.0 mol H_2 at 25 °C; the other contains 1.0 mol N_2 at 25 °C. Calculate the Gibbs energy of mixing when the partition is removed. Assume perfect behavior.

Self-test 7.3

Self-test 5.3 Suppose that 2.0 mol H_2 at 2.0 atm and 25°C and 4.0 mol N_2 at 3.0 atm and 25°C are mixed at constant volume. Calculate $\Delta_{\text{mix}}G$. What would be the value of $\Delta_{\text{mix}}G$ had the pressures been identical initially? [−9.7 kJ, −9.5 kJ]



Self-test T.3

H_2 N_2
 initial (2mol, 2atm, V_1) (4mol, 3atm, V_2)

Final mixture $V = V_1 + V_2$

$$P_1 V_1 = n_1 R T$$

$$V_1 = \frac{2 \text{ mol}}{2 \text{ atm}} R T = \frac{\text{mol}}{\text{atm}} R T$$

$$P_2 V_2 = n_2 R T$$

$$V_2 = \frac{4 \text{ mol}}{3 \text{ atm}} R T = \frac{4}{3} \frac{\text{mol}}{\text{atm}} R T$$

$$V = V_1 + V_2 = \frac{7}{3} \frac{\text{mol}}{\text{atm}} R T$$

$$\therefore p^{\text{final}} (V_1 + V_2) = (n_1 + n_2) \cdot R T$$

$$p^{\text{final}} \frac{7}{3} \frac{\text{mol}}{\text{atm}} R T = 6 \text{ mol} \times R T$$

$$\therefore p^{\text{final}} = \frac{18}{7} \text{ atm}$$

$$\therefore x_1 = \frac{2}{6} = \frac{1}{3}$$

$$x_2 = \frac{4}{6} = \frac{2}{3}$$

$$p_1^{\text{final}} = \frac{18}{7} \text{ atm} \times \frac{1}{3} = \frac{6}{7} \text{ atm}$$

$$p_2^{\text{final}} = \frac{18}{7} \text{ atm} \times \frac{2}{3} = \frac{12}{7} \text{ atm}$$

$$G_{\text{initial}} = \sum \mu_i \cdot n_i \quad \text{加合公式}$$

$$= n_1 \mu_1 + n_2 \mu_2$$

$$= 2 \text{ mol} \times (\mu_1^\ominus + R T \ln p_1) + 4 \text{ mol} (\mu_2^\ominus + R T \ln p_2)$$

$$G_{\text{final}} = n_1 \mu_1^{\text{final}} + n_2 \mu_2^{\text{final}}$$

$$= 2 \text{ mol} (\mu_1^\ominus + R T \ln p_1^{\text{final}}) + 4 \text{ mol} (\mu_2^\ominus + R T \ln p_2^{\text{final}})$$

$$\Delta G_{\text{mix}} = G_{\text{final}} - G_{\text{initial}}$$

$$= 2 \text{ mol} \times R T \ln \frac{p_1^{\text{final}}}{p_1} + 4 \text{ mol} \times R T \ln \frac{p_2^{\text{final}}}{p_2}$$

$$= 2 \text{ mol} \times R T \times \left(\ln \frac{6}{2} + 2 \times \ln \frac{12}{3} \right)$$

$$= 2 \text{ mol} \times 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \times 298 \text{ K} \times \left(\ln \frac{3}{1} + 2 \ln \frac{4}{1} \right)$$

$$= -9.7 \text{ kJ}$$

如果是等压混合

$$\Delta G_{\text{mix}} = n R T (x_1 \ln x_1 + x_2 \ln x_2)$$

$$= 6 \text{ mol} \times 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \times 298 \text{ K} \left(\frac{1}{3} \ln \frac{1}{3} + \frac{2}{3} \ln \frac{2}{3} \right)$$

$$= 6 \times 8.314 \times 298 (-0.3662 - 0.2703)$$

$$= -9.46 \text{ kJ}$$



7.12 a

5.8(a) Consider a container of volume 5.0 dm^3 that is divided into two compartments of equal size. In the left compartment there is nitrogen at 1.0 atm and 25°C ; in the right compartment there is hydrogen at the same temperature and pressure. Calculate the entropy and Gibbs energy of mixing when the partition is removed. Assume that the gases are perfect.

7.13a

5.9(b) Calculate the Gibbs energy, entropy, and enthalpy of mixing when $1.00 \text{ mol C}_6\text{H}_{14}$ (hexane) is mixed with $1.00 \text{ mol C}_7\text{H}_{16}$ (heptane) at 298 K ; treat the solution as ideal.

等压混合

$$\Delta G_{mix} = nRT(x_A \ln x_A + x_B \ln x_B)$$

$$\Delta S_{mix} = -\frac{\Delta G_{mix}}{T} \quad \Delta H_{mix} = \Delta G_{mix} + T \times \Delta S_{mix}$$

5.8(a) Consider a container of volume 5.0 dm^3 that is divided into two compartments of equal size. In the left compartment there is nitrogen at 1.0 atm and 25°C ; in the right compartment there is hydrogen at the same temperature and pressure. Calculate the entropy and Gibbs energy of mixing when the partition is removed. Assume that the gases are perfect.

5.8(a) At constant pressure.

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

$$\because T_1 = T_2 \quad P_1 = P_2 \quad V_1 = V_2 \quad \therefore n_1 = n_2$$

$$\therefore x_A = 0.5 \quad x_B = 0.5$$

$$nRT = PV = 1 \text{ atm} \times 5 \text{ dm}^3 = 5.05 \times 10^2 \text{ J}$$

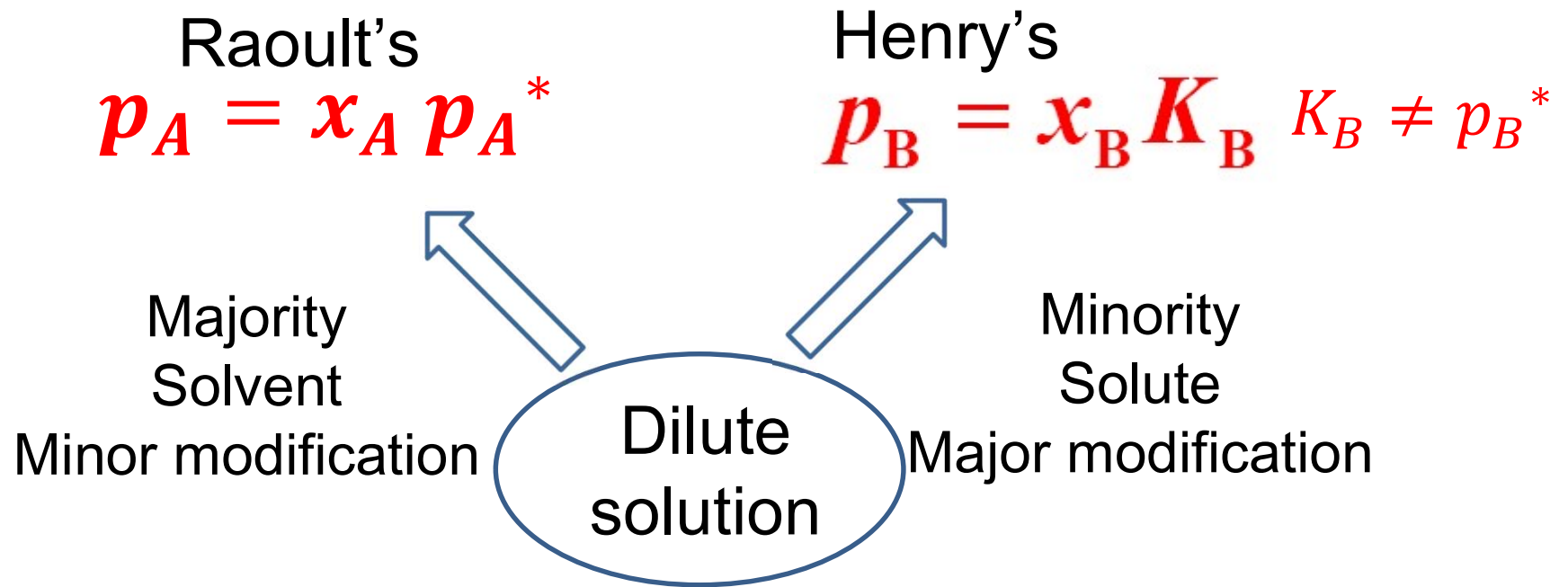
$$\therefore \Delta_{\text{mix}}G = -350 \text{ J}$$

$$\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T} \right)_P = -\Delta_{\text{mix}}G - \Delta_{\text{mix}}S$$

$$\begin{aligned} \therefore \Delta_{\text{mix}}S &= -nR(x_A \ln x_A + x_B \ln x_B) \\ &= 1.17 \text{ J/K} \end{aligned}$$



3. Raoult's law and Henry's law [Self-test 7.4 and Exercise 7.15](#)



Ideal solutions: Mixtures that obey the **Raoult's law** throughout the composition range from pure A to pure B.

Ideal-dilute solutions: Mixtures for which the **solute obeys Henry's law** and the **solvent obeys Raoult's law**.



Exercises 7.15

Use Henry's law and the data in Table 5.1 to calculate the solubility (as a molality) of CO_2 in water at 25°C when its partial pressure is (a) 0.10 atm, (b) 1.00 atm.



$$p_B = x_B K_B$$

K: Pa $\rightarrow$ x molar fraction

K: Pa kg mol⁻¹ $\rightarrow$ molality

$$K_B(CO_2) = 1.67 \times 10^6 \text{ Pa}$$

$$x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} = \frac{n_b M_A}{n_A M_A} = b M_A$$

$$p_B = 1.013 \times 10^4 \text{ Pa}$$

$$x_B = p_B / K_B = 6.07 \times 10^{-3}$$

$$b = x_B / M_A = x_B / 0.018 \text{ kg mol}^{-1} = 3.4 \times 10^{-3} \text{ mol kg}^{-1}$$



$$p_B = b_B K_B$$

$$K_B(CO_2) = 3.01 \times 10^3 \text{ kPa kg mol}^{-1}$$

K: Pa $\rightarrow$ x molar fraction

K: Pa kg mol⁻¹ $\rightarrow$ molality

$$p_B = 1.013 \times 10^4 \text{ Pa}$$

$$\begin{aligned} b_B &= \frac{p_B}{K_B} = \frac{1.013 \times 10^4 \text{ Pa}}{3.01 \times 10^6 \text{ Pa kg mol}^{-1}} \\ &= 3.4 \times 10^{-3} \text{ mol kg}^{-1} \end{aligned}$$



4. Colligative properties Exercises 7.10 and 7.17

- Vapour pressure is always lower

$$p_A = x_A p_A^*$$

$$b = \frac{x_{\text{solute}}}{M_{\text{solvent}}}$$

- Boiling point is always higher

$$\Delta T = K_b b$$

- Freezing point is always lower

$$\Delta T = K_f b$$

- Osmotic pressure drives solvent from lower concentration to higher concentration

$$\Pi = \frac{n_B}{V} RT$$



Exercises 7.10

The addition of 100 g of a compound to 750 g of CCl_4 lowered the freezing point of the solvent by 10.5 K. Calculate the molar mass of the compound.



$$\Delta T = K_f b$$

$$b = \frac{100 \text{ g} / M \text{ g mol}^{-1}}{0.75 \text{ kg}} = \frac{100}{0.75 M}$$

$$K_f(\text{CCl}_4) = 30 \text{ K kg mol}^{-1}$$

$$\frac{100}{0.75 M} \times 30 \text{ K kg mol}^{-1} = 10.5 \text{ K}$$

$$M = 380.95 \text{ g mol}^{-1}$$



5. Activity

➤ Ideal-dilute solutions

$$x_A = \frac{p_A}{p_A^*} \Rightarrow \text{Raoult's law} \quad \text{solvent}$$

$$x_B = \frac{p_B}{K} \Rightarrow \text{Henry's law} \quad \text{solute}$$

➤ Real solvent

$$x_A \neq \frac{p_A}{p_A^*} = \gamma_A x_A = a_A$$

a : the activity

➤ Real solute

$$x_B \neq \frac{p_B}{K_B} = \gamma_B x_B = a_B$$

γ : the activity coefficient



7.20b

5.15(b) Given that $p^*(\text{H}_2\text{O}) = 0.02308 \text{ atm}$ and $p(\text{H}_2\text{O}) = 0.02239 \text{ atm}$ in a solution in which 0.122 kg of a non-volatile solute ($M = 241 \text{ g mol}^{-1}$) is dissolved in 0.920 kg water at 293 K, calculate the activity and activity coefficient of water in the solution.

A: water

B: solute

$$a_A = \frac{p_A}{p_A^*} = \frac{0.02239}{0.02308} = 0.9701$$

$$n_A = \frac{m_A}{M_A} = \frac{920\text{g}}{18\text{g mol}^{-1}} = 51.1\text{mol}$$

$$n_B = \frac{m_B}{M_B} = \frac{112\text{g}}{241\text{g mol}^{-1}} = 0.465 \text{ mol}$$

$$x_A = 0.989$$

$$\gamma_A = \frac{a_A}{x_A} = \frac{0.9701}{0.989} = 0.981$$



6. Chemical potential

$$\mu = \mu^{\ominus} + RT \ln a$$



$a = 1$
 a 无量纲

$$\mu = \mu^{\ominus}$$

所以standard state首先是 $a=1$ 的状态



纯气体

$$\mu = \mu^{\ominus}(T) + RT \ln \frac{p}{p^{\ominus}}$$

standard state: $\frac{p}{p^{\ominus}} = 1$

pure gas, $T, p = p^{\ominus} = 1 \text{ bar}$

混合气体

$$\mu_B = \mu_B^{\ominus}(T) + RT \ln \frac{p_B}{p^{\ominus}}$$

standard state: $\frac{p_B}{p^{\ominus}} = 1$

pure gas B, $T, p = p^{\ominus} = 1 \text{ bar}$



理想混合气体中某种气体的分压等于这种气体单独占据整个体积时的压力



溶液以溶剂为观测对象

$$\mu_A = \mu_A^\ominus + RT \ln a_A = \mu_A^\ominus + RT \ln \gamma_A x_A$$

溶液越稀薄，溶剂的比例越高，溶液越接近理想溶液，符合Raoult's law

$$\lim_{x_A \rightarrow 1} \gamma_A = 1$$

standard state:

$$a_A = 1$$

pure liquid A, T, p[⊖]



溶液以溶质为观测对象

$$\mu_B = \mu_B^\ominus + RT \ln a_B = \mu_B^\ominus + RT \ln \gamma_B \frac{b_B}{b^\ominus}$$

溶液越稀薄，溶液越接近理想稀薄溶液，符合Henry's law

$$\lim_{b_B \rightarrow 0} \gamma_B = 1$$

standard state:

$$a_B = 1$$

$$\gamma_B = 1 \quad b_B = b^\ominus = 1 \text{ mol kg}^{-1}$$

$b_B = 1 \text{ mol kg}^{-1}$, T, p obeys Henry's law,

Hypothetical state

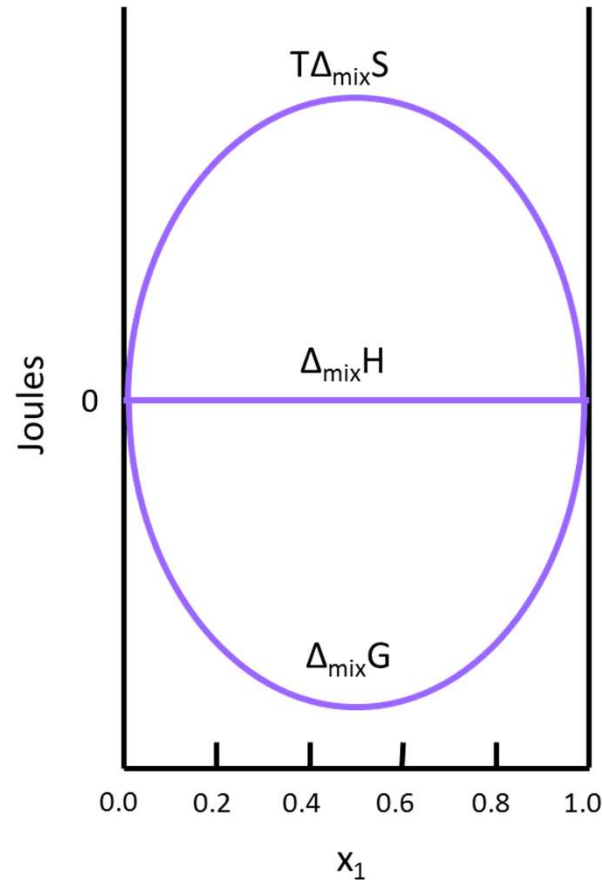
因此 $a=1$ 的状态是远离Henry's law的状态，因此我们假想在 $a=1$ 的状态下仍然符合Henry's law



When argon is placed in a container of neon, the argon spontaneously disperses throughout the neon because _____.

- A) of hydrogen bonding
- B) of solvent-solute interactions
- C) a decrease in energy occurs when the two mix
- D) the dispersion of argon atoms produces an increase in disorder
- E) of the large attractive forces between argon and neon atoms

D



$$\Delta_{\text{mix}}S > 0$$

$$\Delta_{\text{mix}}H = 0$$

$$\Delta_{\text{mix}}G < 0$$

Figure 2. A graph of $T\Delta_{\text{mix}}S$, $\Delta_{\text{mix}}H$, and $\Delta_{\text{mix}}G$ as functions of x_1 for the mixing of two ideal gases.



Chapter 5

Physical Transformations of Pure Substances

Chapter 7

Phase diagrams



Summary

1. One-component Phase diagrams

□ $T_{\text{trs}}: T_b, T_f, T_{\text{sub}}, T_3, T_c$

□ The stability of phase

Phase equilibrium: $\mu(\alpha) = \mu(\beta)$

Phase transition: $\mu(\alpha) > \mu(\beta)$

Only one phase, an area:

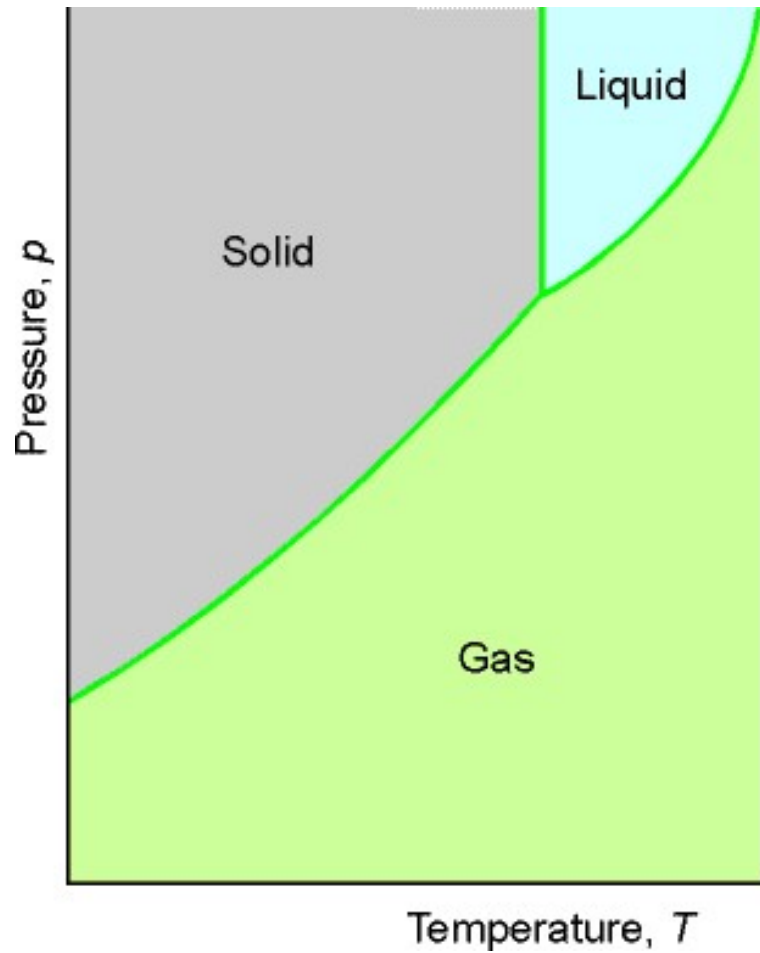
$$C=1, P=1 \quad \Rightarrow \quad F=1-1+2=2$$

Two phases, The lines:

$$C=1, P=2 \quad \Rightarrow \quad F=1-2+2=1$$

three phases, A point:

$$C=1, P=3 \quad \Rightarrow \quad F=1-3+2=0$$



$$\left(\frac{\partial \mu}{\partial T}\right)_p = -S_m$$

$$\left(\frac{\partial \mu}{\partial p}\right)_T = V_m$$



□ Phase boundary---Clapeyron equation

Exercise 6.4, 6.5, 6.8, 6.12

$$\frac{dp}{dT} = \frac{\Delta_{trs}S}{\Delta_{trs}V} = \frac{\Delta_{trs}H}{T\Delta_{trs}V}$$

注意相变方向的一致性

Liquid-vapour boundary

$$\Delta p \rightarrow \Delta T$$



- Vapour pressure

$$p = p^* e^{V_m \Delta P / RT}$$
$$\frac{p - p^*}{p^*} \approx \frac{V_m \Delta P}{RT}$$

外压变化和蒸汽压的关系
The relationship between the external pressure and vapor pressure

$$p = p^* e^{-\chi}$$
$$\chi = \frac{\Delta_{vap} H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

已知一个温度下的蒸汽压求另外一个温度下的蒸汽压（温度和蒸汽压的关系）
The relationship between temperature and vapor pressure

6.4 (b) The vapour pressure of a substance at 20.0°C is 58.0 kPa and its enthalpy of vaporization is 32.7 kJ mol^{-1} . Estimate the temperature at which its vapour pressure is 66.0 kPa .

$$\begin{array}{ll} 6.4b. & P_1 = 5.8 \times 10^4 \text{ Pa} \quad T_1 = 293 \text{ K} \\ & P_2 = 6.6 \times 10^4 \text{ Pa} \quad \Delta_{\text{vap}}H = 32.7 \text{ kJ/mol} \quad T_2 = ? \end{array}$$

$$P_2 = P_1 e^{-x}$$

$$6.6 \times 10^4 = 5.8 \times 10^4 \cdot e^{-x} \Rightarrow x = -0.13$$

$$x = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$-0.13 = \frac{32.7 \times 10^3}{8.314} \left(\frac{1}{T_2} - \frac{1}{293} \right)$$

$$T_2 = 296 \text{ K} \Rightarrow 23^{\circ}\text{C}.$$

6.5 (a) The molar volume of a certain solid is $161.0 \text{ cm}^3 \text{ mol}^{-1}$ at 1.00 atm and 350.75 K, its melting temperature. The molar volume of the liquid at this temperature and pressure is $163.3 \text{ cm}^3 \text{ mol}^{-1}$. At 100 atm the melting temperature changes to 351.26 K. Calculate the enthalpy and entropy of fusion of the solid.

6.5a. $V_m(s) = 161 \text{ cm}^3/\text{mol} = 161 \times 10^{-6} \text{ L/mol}$

$V_m(l) = 163.3 \text{ cm}^3/\text{mol} = 163.3 \times 10^{-6} \text{ L/mol}$

$P_1 = 1 \text{ atm} \quad T_1 = 350.75 \text{ K}$

$P_2 = 100 \text{ atm} \quad T_2 = 351.26 \text{ K}$

$$\frac{\Delta P}{\Delta T} = \frac{\Delta_{\text{fus}} S}{\Delta_{\text{fus}} V} = \frac{\Delta_{\text{fus}} H}{T \Delta_{\text{fus}} V}$$

↓
变化. 求 S.

$$\frac{99 \times 1.013 \times 10^5 \text{ Pa}}{351.26 - 350.75 \text{ K}} = \frac{\Delta_{\text{fus}} S}{(163.3 - 161) \times 10^{-6} \text{ L/mol}}$$

$$\Delta_{\text{fus}} S = 45.1 \text{ J/mol} \cdot \text{K}$$

$$\begin{aligned} \Delta_{\text{fus}} H &= T_1 \cdot \Delta_{\text{fus}} S = 350.75 \times 45.1 \text{ J/mol} \cdot \text{K} \\ &= 15.8 \text{ kJ/mol} \end{aligned}$$



Exercises 6.8a

When benzene freezes at 5.5°C its density changes from 0.879 g cm^{-3} to 0.891 g cm^{-3} . Its enthalpy of fusion is $10.59 \text{ kJ mol}^{-1}$. Estimate the freezing point of benzene at 1000 atm.

	Solid	liquid
$\Delta_{trs}V \rightarrow \rho \text{ (g m}^{-3}\text{)}$	8.91×10^5	8.79×10^5



$$\Delta_{trs}H = 1.059 \times 10^4 \text{ J mol}^{-1}$$



M (benzene)=78 g mol⁻¹

$$\frac{dp}{dT} = \frac{\Delta_{fus}H}{T\Delta_{fus}V} = \frac{\Delta_{fus}H}{T(V_m(l) - V_m(s))} = \frac{\Delta_{fus}H}{TM\left(\frac{1}{\rho_l} - \frac{1}{\rho_s}\right)}$$

$$\frac{dp}{dT} = \frac{1.059 \times 10^4 \text{ J mol}^{-1}}{278.5 \text{ K} \times 78 \text{ g mol}^{-1} \times \left(\frac{1}{8.79 \times 10^5 \text{ g m}^{-3}} - \frac{1}{8.91 \times 10^5 \text{ g m}^{-3}}\right)}$$

$$\frac{dp}{dT} \approx \frac{\Delta p}{\Delta T} = \frac{(1000 - 1) \times 1.013 \times 10^5 \text{ Pa}}{T_f - 278.5} = 0.325 \times 10^8 \text{ J m}^{-3} \text{ K}^{-1}$$

$$T_f = 281.6 \text{ K}$$

The boiling point of hexane is 69.0°C . Estimate (a) its enthalpy of vaporization and (b) its vapour pressure at 25°C and 60°C .

6.11 b. According to Trouton's constant $\frac{\Delta_{\text{vap}}H}{T} = 85 \text{ J/K}\cdot\text{mol}$

$$T_b = 342 \text{ K}$$

$$\Delta_{\text{vap}}H = 85 \text{ J/K}\cdot\text{mol} \times 342 \text{ K} = 29.07 \text{ kJ/mol}$$

$$\ln \frac{P_2}{P_1} = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\therefore T_1 = 342 \text{ K} \quad P_1 = 1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$

$$\frac{P_2}{P_1} = e^{\chi} \quad \chi = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\chi_a = \frac{29.07 \times 10^3}{8.314} \left(\frac{1}{342} - \frac{1}{298} \right)$$

$$= -1.51$$

$$P_2 = 0.22 \text{ atm}$$

$$\chi_b = \frac{29.07 \times 10^3}{8.314} \left(\frac{1}{342} - \frac{1}{333} \right)$$

$$= -0.276$$

$$P_3 = 0.76 \text{ atm}$$



Exercises 6.12b

Calculate the melting point of ice under a pressure of 10 MPa. Assume that the density of ice under these conditions is approximately 0.915 g cm^{-3} and that of liquid water is 0.998 g cm^{-3} .

	Solid	liquid
$\Delta_{trs}V \rightarrow \rho \text{ (g m}^{-3}\text{)}$	9.15×10^5	9.98×10^5



$$\Delta_{fus}H = 6.008 \times 10^3 \text{ J mol}^{-1}$$

From table 2.3



M (benzene)=18 g mol⁻¹

$$\frac{dp}{dT} = \frac{\Delta_{fus}H}{T\Delta_{fus}V} = \frac{\Delta_{fus}H}{T(V_m(l) - V_m(s))} = \frac{\Delta_{fus}H}{TM(\frac{1}{\rho_l} - \frac{1}{\rho_s})}$$

$$\frac{dp}{dT} = \frac{6.008 \times 10^3 J mol^{-1}}{273K \times 18 g mol^{-1} \times (\frac{1}{9.98 \times 10^5 g m^{-3}} - \frac{1}{9.15 \times 10^5 g m^{-3}})}$$

$$\frac{dp}{dT} \approx \frac{\Delta p}{\Delta T} = \frac{(100 - 1) \times 10^5 Pa}{T_f - 273} = -1.344 \times 10^7 J m^{-3} K^{-1}$$

$$T_f = 272.26 K$$



2. Gibbs phase rule [Exercise 8.10 and 8.11](#)

$$F = C - P + 2$$

F: the degree of freedom or variance;

C: the number of components;

P: the number of phases at equilibrium.

$$C = S - R - R'$$

S--- A constituent is a chemical species that is present in a system

R---the number of independent reactions

R'---the number of concentration limitation (in one phase)



6.7(b) Ammonium chloride, NH_4Cl , decomposes when it is heated.

(a) How many components and phases are present when the salt is heated in an otherwise empty container? (b) Now suppose that additional ammonia is also present. How many components and phases are present?



(a) $P=2$; $C=1$

(b) $P=2$; $C=2$



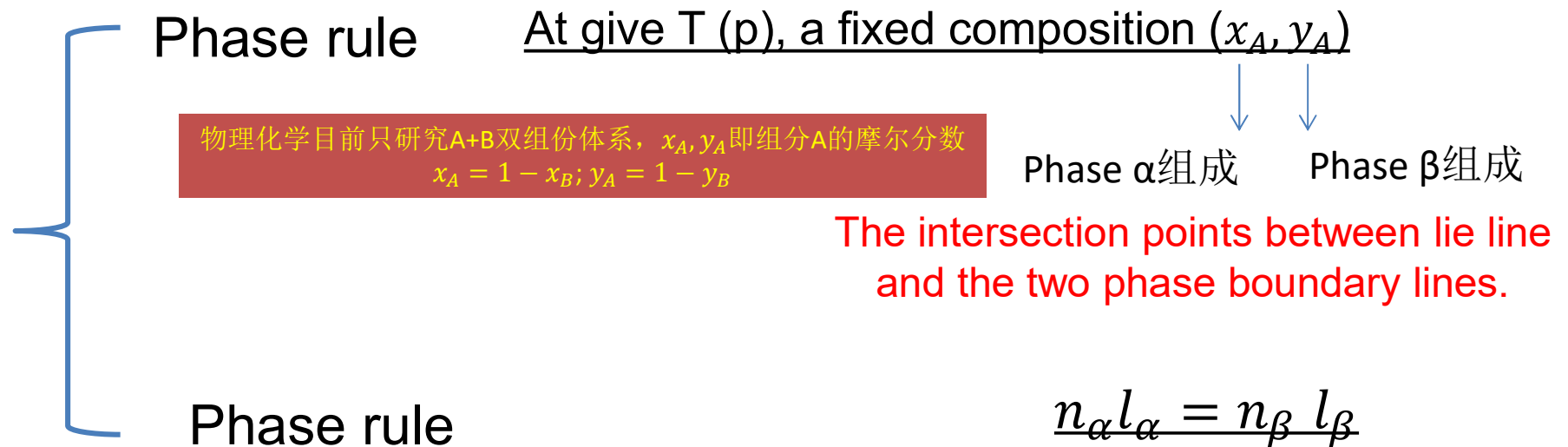
- Phase rule

$$F = C - P + 2$$

- The interesting situation: $F=1$ in two-Component phase diagram

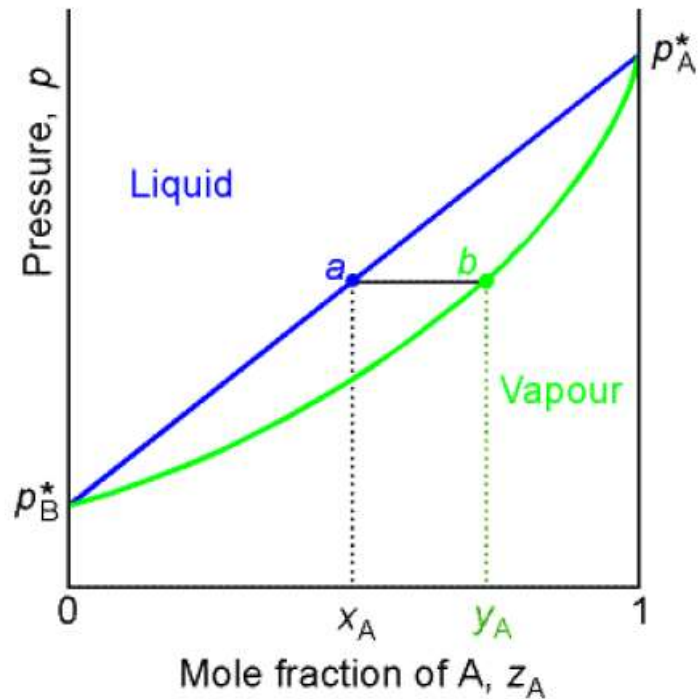
物理意义: a two phases co-existing area

外观: The area surrounded by two phase boundaries.

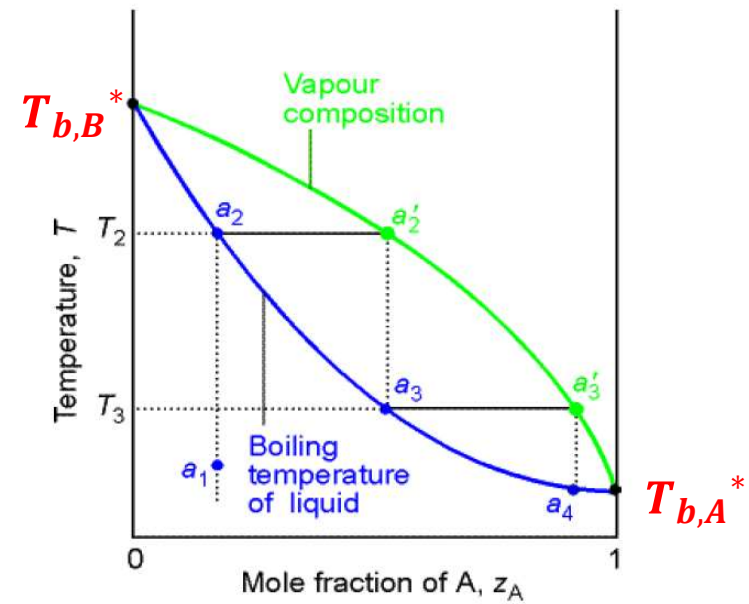




Vapour pressure-composition diagrams



Temperature-composition diagrams





Vapour pressure-composition diagrams

$$p_A = x_A p_A^*$$

Exercise 8.4 and 8.6

$$p_B = x_B p_B^*$$

$$p = p_A + p_B = x_A p_A^* + x_B p_B^* = p_B^* + (p_A^* - p_B^*) x_A$$

Liquid phase composition (x_A, x_B)

Raoult's law

$$x_A = \frac{p_A}{p_A^*} \quad x_B = 1 - x_A = \frac{p_B}{p_B^*}$$

Vapour phase composition (y_A, y_B)

Dalton's law

$$y_A = \frac{p_A}{p} \quad y_B = 1 - y_A = \frac{p_B}{p}$$



3. Two-component phase diagrams

- Miscible ideal solutions [Figure 8.14](#)

Vapour pressure-composition diagrams

Temperature-composition diagrams

The Level rule on the tie line

- Miscible real solution [Figure 8.16; 8.17](#)

Azeotropic point and azeotrope

Distillation

- Immiscible liquids

- Partially miscible liquid-liquid diagrams

Critical point [Figure 8.19; 8.24; 8.25; 8.26; 8.27](#)
[Example 8.3](#)

Vapour-liquid
Phase transition

Application

Distillation



Exercises 8.6a

It is found that the boiling point of a binary solution of A and B with $x_A = 0.6589$ is 88°C . At this temperature the vapour pressures of pure A and B are 127.6 kPa and 50.60 kPa, respectively. (a) Is this solution ideal? (b) What is the initial composition of the vapour above the solution?



$$p_A^* = 127.6 \text{ kPa} \quad x_A = 0.6589$$

$$p_B^* = 50.6 \text{ kPa} \quad x_B = 0.3411$$

$$p_A = x_A p_A^* = 84.0756 \text{ kPa}$$

$$p_B = x_B p_B^* = 17.2597 \text{ kPa}$$

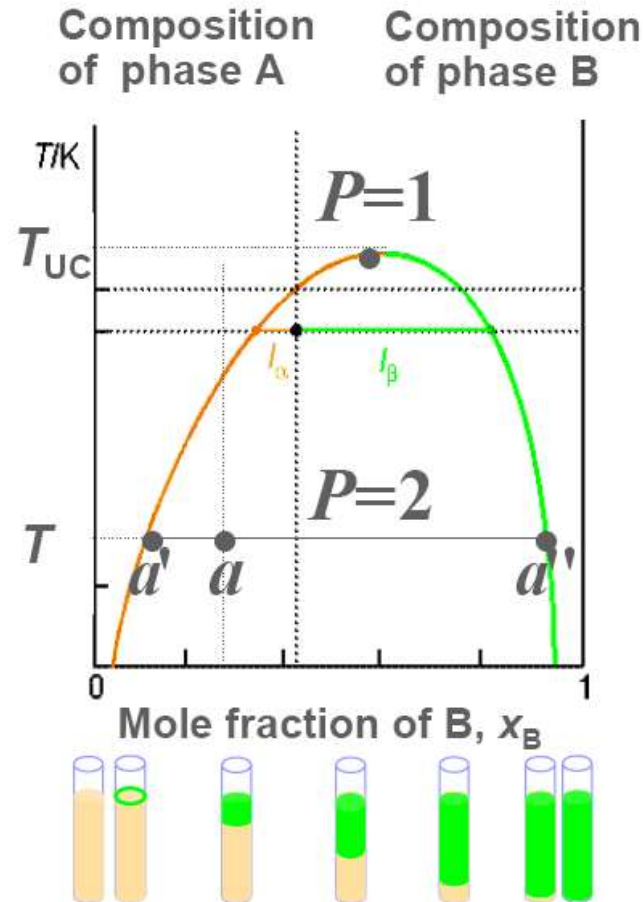
$$p = p_A + p_B = 101.3 \text{ kPa}$$

The results obey Raoult's law and therefore the solution is ideal .

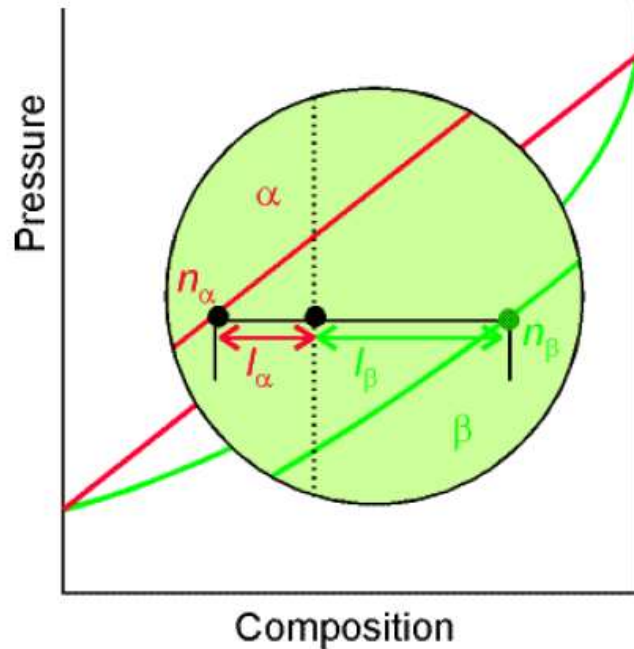
$$y_A = \frac{p_A}{p} = 0.83 \quad y_B = 1 - y_A = 0.17$$



Partially miscible liquid-liquid diagrams



The T - C diagram for hexane and nitrobenzene at 1 atm



Example 8.2

The level rule

$$n_{\alpha}l_{\alpha} = n_{\beta}l_{\beta}$$

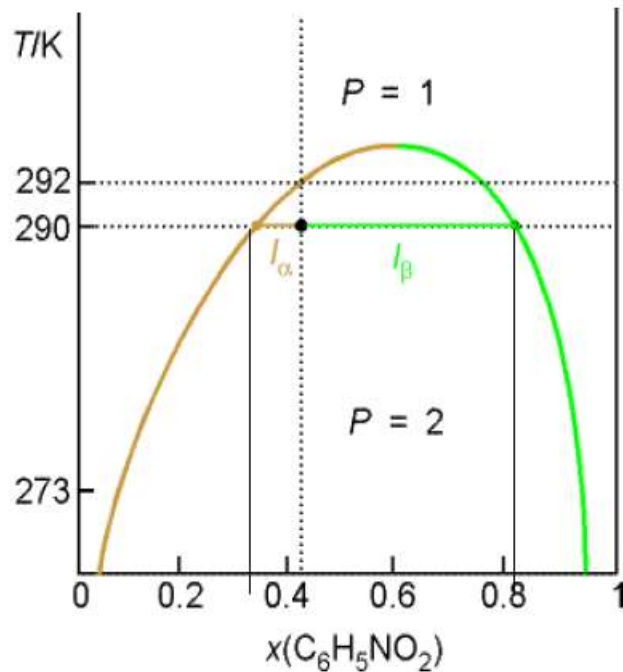
The points on the tie line have the same compositions (x_A, y_A).

However, the molar ratio between the vapour phase and liquid phase varied according to the level rule.



Interpreting a liquid-liquid phase diagram

A mixture of 50 g of hexane (0.59 mol C_6H_{14}) and 50 g nitrobenzene (0.41 mol $\text{C}_6\text{H}_5\text{NO}_2$) was prepared at 290 K. What are the compositions of the phases, and in what proportions do they occur? To what temperature must the sample be heated in order to obtain a single phase?



T-C diagram for hexane and nitrobenzene at 1 atm.

Answer: denote hexane by H and nitrobenzene by N;

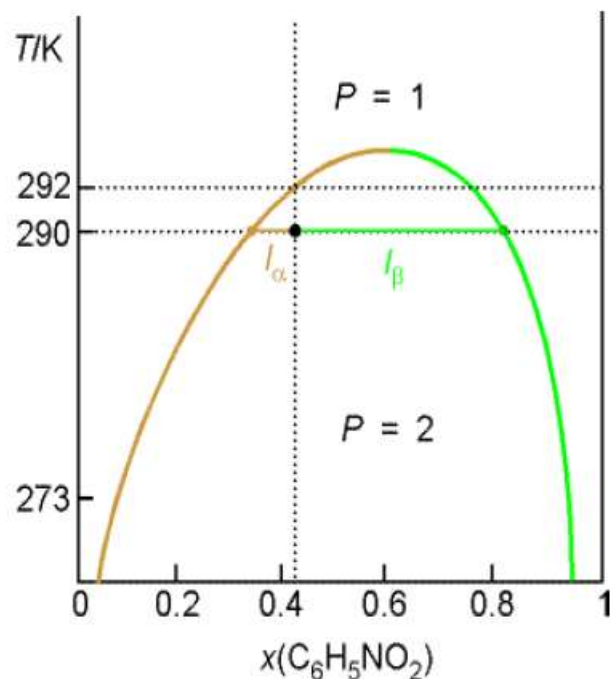
The point $x_H = 0.41$, $T = 290$ K occurs in the two-phase region of the phase diagram.

(1) The compositions of the two phases

The horizontal tie line cuts the phase boundary at

$x_H = 0.35$ and $x_N = 0.83$,

so those are the compositions of the two phases.



T-C diagram for hexane and nitrobenzene at 1 atm.

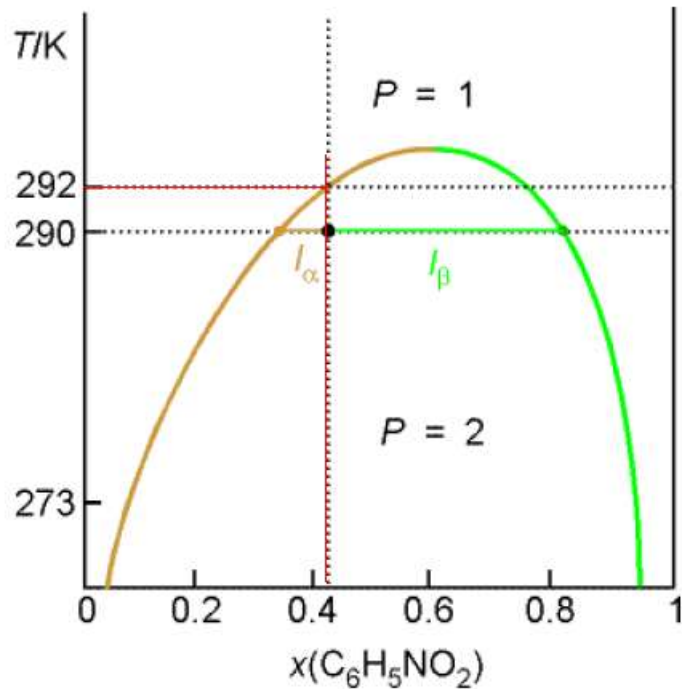
(2). The proportion of the mixture at $T=290\text{K}$

$$n_{\alpha}l_{\alpha} = n_{\beta}l_{\beta}$$

so the ratio of the amounts of each phase is equal to the ratio of the distances l_{α} and l_{β}

$$l_{\alpha} = 0.41 - 0.35; \quad l_{\beta} = 0.83 - 0.41$$

$$\frac{n_{\alpha}}{n_{\beta}} = \frac{l_{\beta}}{l_{\alpha}} = \frac{0.83 - 0.41}{0.41 - 0.35} = 7$$



***T-C* diagram for hexane and nitrobenzene at 1 atm.**

(3). to T obtain a single phase

Heating the sample to

$T=292\text{ K}$

takes into the single-phase region.



- Liquid-solid phase diagrams

The cooling curves

Eutectics

[Figure 8.29; 8.31; 8.32](#)

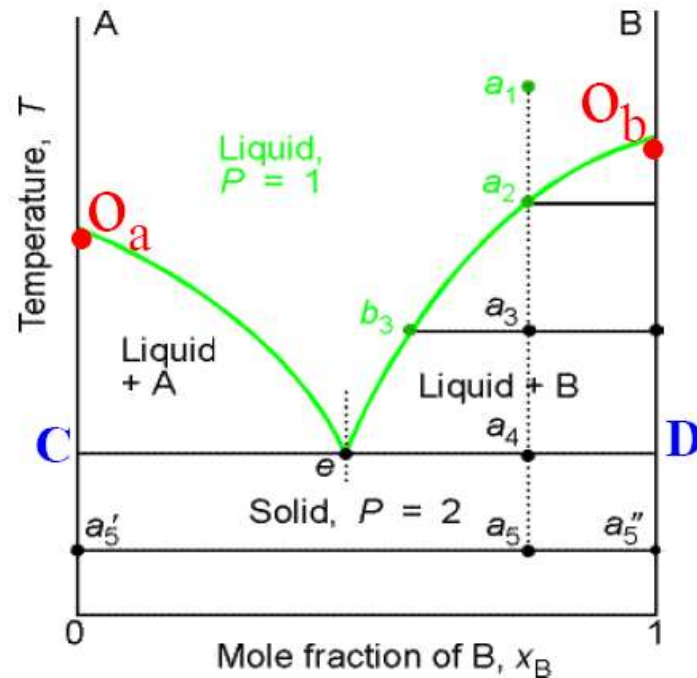


Application

Alloy



Binary eutectic alloy system



The temperature-composition phase diagram for two almost immiscible solids and their completely miscible liquids.

The isopleth through e corresponds to the **eutectic** composition, the mixture with lowest melting point.

Points, lines, and regions

O_a : freezing point of pure a

O_b : freezing point of pure b

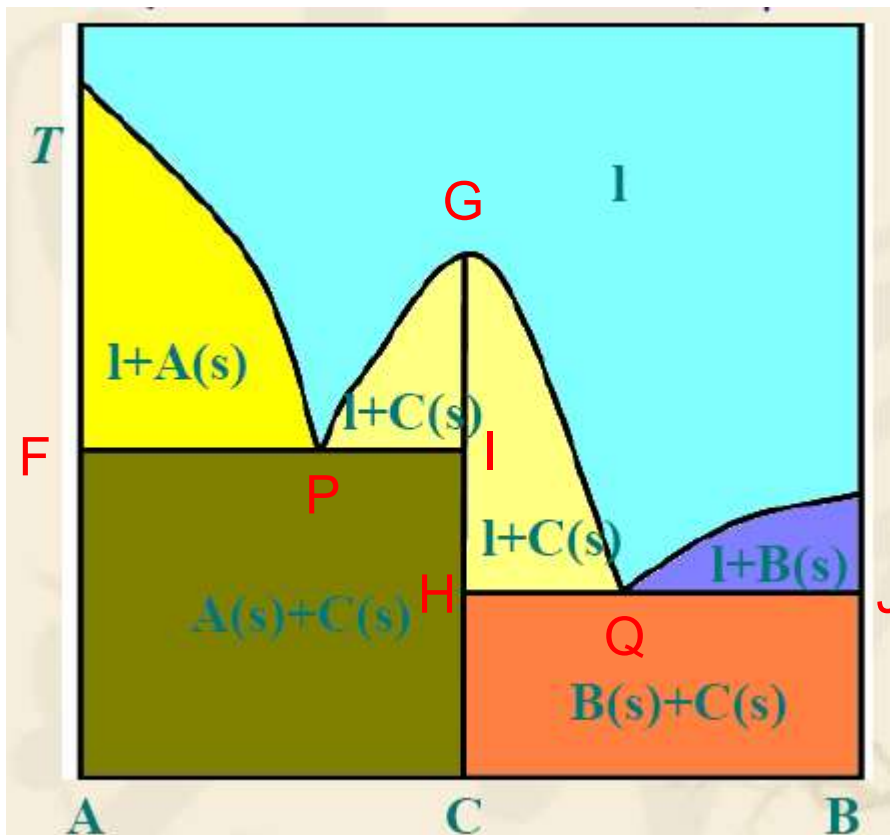
O_a-e, O_b-e : liquid phase line

O_a-C, O_b-D : solid phase line

$C-e-D$: three phase line



$A + B = C$ (stable both in the liquid and solid phase)
 $C=AB$



Two triple lines:

FI: $P=3$, (l, A, C) , $F=0$

JH: $P=3$, (l, C, B) , $F=0$

P: Eutectic mixture of A and C

Q: Eutectic mixture of B and C

G: the melt point of compound C

Calculate the change in the chemical potential of a perfect gas when it expands isothermally at a temperature of 20.0°C so that its volume doubles.

- a) $-1.69 \text{ kJ mol}^{-1}$
- b) -115 J mol^{-1}
- c) $+1.69 \text{ kJ mol}^{-1}$
- d) $+115 \text{ J mol}^{-1}$

13) At 20 C and 1 atm, which of the following is true?

- a) The chemical potential of liquid water is greater than that of either ice or water vapor.
- b) The chemical potential of liquid water is greater than that of ice but less than that of water vapor.
- c) **The chemical potential of liquid water is less than that of either ice or water vapor.**
- d) The chemical potential of liquid water is less than that of ice but greater than that of water vapor.



南方科技大学

谢谢大家的努力

祝考试顺利！

下学期再见！